

CLASS ACTIVITY

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EXP1:- Daily Sales Time Series — R

CODE:-

```
set.seed(10)

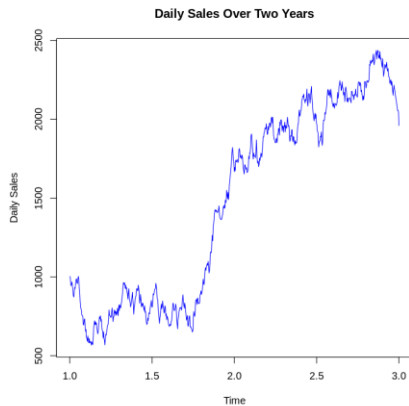
date <- seq(as.Date("2024-01-01"),
            as.Date("2025-12-31"), by="day")

sales <- 1000 + cumsum(rnorm(length(date), 2, 30))

sales_ts <- ts(sales, frequency=365)

plot(sales_ts,
     main="Daily Sales Over Two Years",
     xlab="Time",
     ylab="Daily Sales",
     col="blue")
```

OUTPUT:-



2. Server Temperature Time Series

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
hours = np.arange(1, 25)
```

```
temperature = [35, 36, 37, 38, 39, 41, 43, 45,
```

```
47, 46, 44, 42, 40, 39, 41, 43,
```

```
46, 48, 51, 53, 52, 49, 45, 42]
```

```
threshold = 50
```

```
plt.plot(hours, temperature, marker='o')
```

```
plt.axhline(threshold, linestyle='--', label='Safe Threshold')
```

```
for i in range(len(hours)):
```

```
    if temperature[i] > threshold:
```

```
        plt.scatter(hours[i], temperature[i], s=80)
```

```
plt.title("Hourly Server Temperature")
```

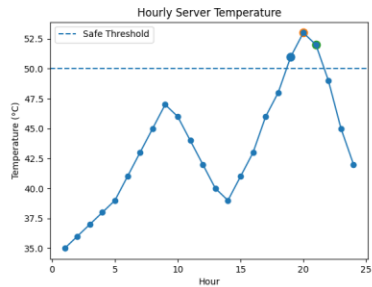
```
plt.xlabel("Hour")
```

```
plt.ylabel("Temperature (°C)")
```

```
plt.legend()
```

```
plt.show()
```

OUTPUT:-



3. Three Tech Stocks — R

CODE:-

```
set.seed(20)
```

```
date <- seq(as.Date("2025-01-01"),
```

```
          as.Date("2025-12-31"), by="day")
```

```
stockA <- 100 + cumsum(rnorm(length(date), 0.2, 2))
```

```
stockB <- 100 + cumsum(rnorm(length(date), 0.3, 1))
```

```
stockC <- 100 + cumsum(rnorm(length(date), 0.2, 3))
```

```
plot(date, stockA,
```

```
     type="l", col="blue",
```

```
     xlab="Date", ylab="Closing Price",
```

```
     main="Tech Stock Price Comparison")
```

```
lines(date, stockB, col="red")
```

```
lines(date, stockC, col="green")
```

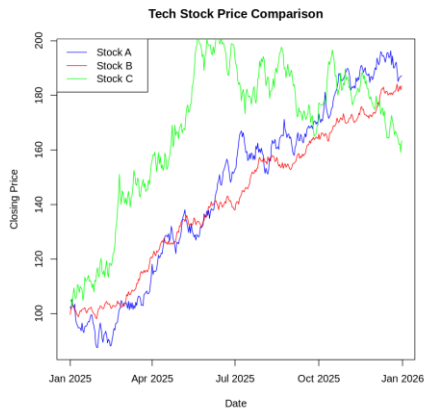
```
legend("topleft",
```

```

legend=c("Stock A","Stock B","Stock C"),
col=c("blue","red","green"),
lty=1)

```

OUTPUT:-



4. Monthly Electricity Consumption

CODE:-

```

import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]

north = [320, 350, 330, 400, 360, 380]

south = [300, 310, 350, 340, 390, 370]

east = [280, 400, 300, 450, 320, 420]

west = [350, 360, 370, 380, 390, 400]

plt.plot(months, north, marker='o', label="North")
plt.plot(months, south, marker='o', label="South")
plt.plot(months, east, marker='o', label="East")
plt.plot(months, west, marker='o', label="West")

```

```
plt.title("Monthly Electricity Consumption")
```

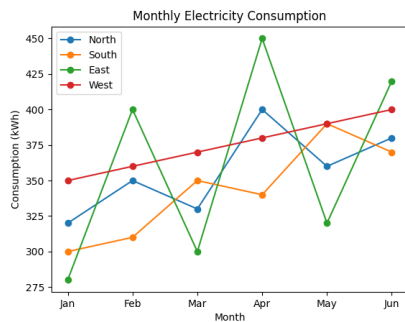
```
plt.xlabel("Month")
```

```
plt.ylabel("Consumption (kWh)")
```

```
plt.legend()
```

```
plt.show()
```

OUTPUT:-



5. Drug Dose–Response Curve — R

```
dose <- c(1, 2, 4, 8, 16, 32)
```

```
response <- c(10, 20, 38, 62, 78, 82)
```

```
plot(dose, response,
```

```
type="o",
```

```
pch=19,
```

```
col="blue",
```

```
main="Drug Dose-Response Curve",
```

```
xlab="Dose",
```

```
ylab="Patient Response")
```

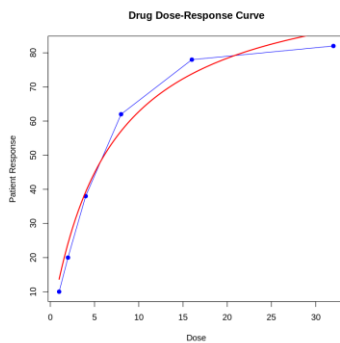
```
model <- nls(
```

```

response ~ a * dose / (b + dose),
start=list(a=100, b=5)
)
curve(
  predict(model, data.frame(dose=x)),
  add=TRUE,
  col="red",
  lwd=2
)

```

OUTPUT:-



6. Fertilizer Dose–Response Curve

CODE:-

```

import numpy as np
import matplotlib.pyplot as plt

fertilizer = np.array([0, 20, 40, 60, 80, 100, 120])
yield_value = np.array([20, 35, 50, 62, 70, 73, 74])

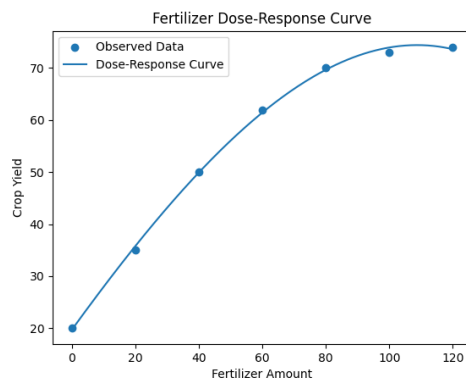
```

```

model = np.polyfit(fertilizer, yield_value, 3)
curve = np.poly1d(model)
x = np.linspace(0, 120, 200)
plt.scatter(fertilizer, yield_value, label="Observed Data")
plt.plot(x, curve(x), label="Dose-Response Curve")
plt.title("Fertilizer Dose-Response Curve")
plt.xlabel("Fertilizer Amount")
plt.ylabel("Crop Yield")
plt.legend()
plt.show()

```

OUTPUT :-



7. STL Decomposition — R

CODE:-

```

sales <- c(
  120,130,150,140,160,180,
  200,190,170,150,140,130,
  125,135,155,145,165,185,

```

205,195,175,155,145,135

)

```
sales_ts <- ts(sales,
```

```
frequency=12,
```

```
start=c(2024,1))
```

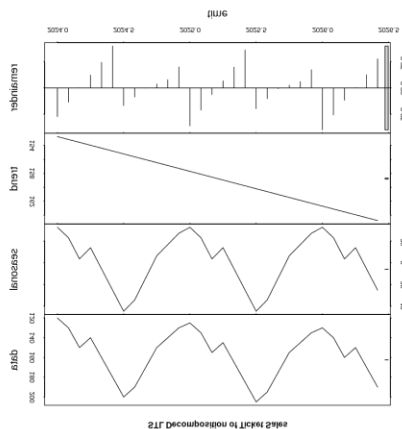
```
plot(
```

```
  stl(sales_ts, s.window="periodic"),
```

```
  main="STL Decomposition of Ticket Sales"
```

```
)
```

OUTPUT:-



8. STL Decomposition — Weekly User Activity

CODE:-

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from statsmodels.tsa.seasonal import STL
```

```
users = [100, 105, 110, 108, 115, 120, 125,
```

```
130, 128, 135, 140, 138, 145, 150,
148, 145, 142, 140, 138, 135, 130,
128, 125, 123, 120, 118, 115, 110]
```

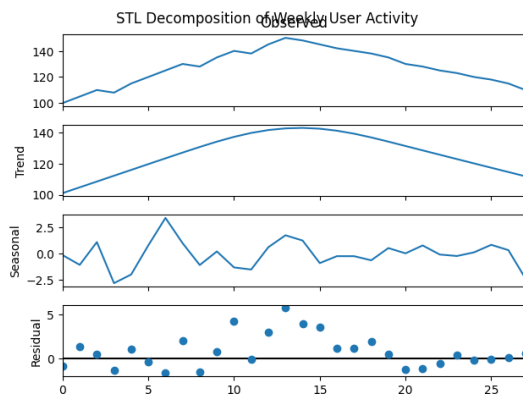
```
result = STL(users, period=7).fit()
```

```
result.plot()
```

```
plt.suptitle("STL Decomposition of Weekly User Activity")
```

```
plt.show()
```

OUTPUT:-



9. 7-Day and 30-Day Moving Average — R

CODE:-

```
set.seed(30)
```

```
price <- 100 + cumsum(rnorm(100, 0.3, 3))
```

```
ma7 <- filter(price, rep(1/7, 7), sides=2)
```

```
ma30 <- filter(price, rep(1/30, 30), sides=2)
```

```
plot(price,
```

```
type="l",
```

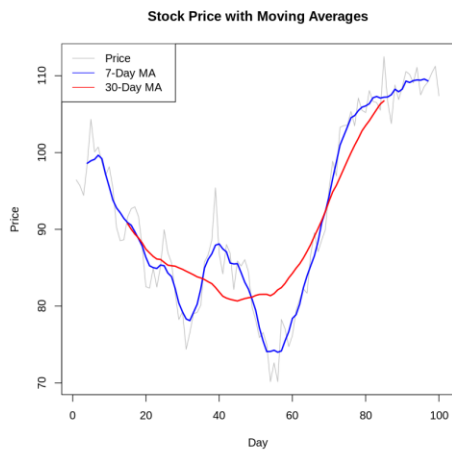
```
col="gray",
```

```

    main="Stock Price with Moving Averages",
    xlab="Day",
    ylab="Price")
lines(ma7, col="blue", lwd=2)
lines(ma30, col="red", lwd=2)
legend("topleft",
      legend=c("Price","7-Day MA","30-Day MA"),
      col=c("gray","blue","red"),
      lty=1)

```

OUTPUT:-



10. GDP Detrending

CODE:-

```

import numpy as np
import matplotlib.pyplot as plt
from scipy import signal
gdp = np.array([

```

100, 103, 106, 108,
112, 115, 118, 121,
125, 127, 130, 134,
138, 140, 143, 147

)

```
detrended = signal.detrend(gdp)
plt.plot(detrended, marker='o')
plt.axhline(0, linestyle='--')
plt.title("Detrended Quarterly GDP")
plt.xlabel("Quarter")
plt.ylabel("Detrended GDP")
plt.show()
```

OUTPUT:-

